

Real Time Water Pipeline Leakage Detection Using IoT

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Abstract—Water leakage in pipeline networks leads to significant water loss and increased maintenance costs. This paper presents a real-time water leakage detection system using Internet of Things (IoT) technology. The system utilizes ESP32, flow sensors, and pressure sensors to monitor water flow and pressure across multiple nodes. Leakage is detected by analyzing variations in flow rate and pressure values. LoRa communication enables long-distance data transmission, while real-time monitoring is achieved using the Blynk mobile application and a web dashboard. The system also provides automatic control by shutting off the solenoid valve when leakage is detected. Experimental results demonstrate that the proposed system accurately detects leakage location and distance with fast response time. The system is cost-effective, reliable, and suitable for smart water management applications.

Keywords—IoT, Real-Time Monitoring, Water Leakage Detection, ESP32, LoRa, Flow Sensor, Pressure Sensor

I. INTRODUCTION

Water is one of the most essential resources for human life, and efficient management of water distribution systems is crucial. A significant amount of water is lost due to leakages in pipelines, which leads to resource wastage, increased operational costs, and environmental impact. Traditional methods of leakage detection are manual, time-consuming, and often inefficient, especially in large-scale pipeline networks.

With the advancement of Internet of Things (IoT) technology, real-time monitoring and automation have become possible. This paper presents a real-time water leakage detection system using IoT, which utilizes ESP32 microcontrollers, flow sensors, and pressure sensors to continuously monitor pipeline conditions. The system detects leakage by analyzing variations in flow rate and pressure at different nodes.

To enable long-distance communication, LoRa technology is used for wireless data transmission between nodes and the central monitoring unit. The collected data is displayed in real time using an LCD, Blynk mobile application, and a web dashboard. Additionally, the system includes an automatic control mechanism that shuts off the solenoid valve when leakage is detected, thereby preventing further water loss.

The proposed system is cost-effective, reliable, and suitable for smart water management applications in industries, agriculture, and urban infrastructure.

II. PROPOSED SYSTEM

The proposed system consists of one central node and multiple remote nodes placed at different points along the water pipeline. Each node is equipped with a flow sensor, pressure sensor, and an ESP32 microcontroller to monitor the flow rate and pressure of water in real time.

The remote nodes collect sensor data and transmit it to the central node using LoRa communication, which enables long-distance wireless data transfer. The central node receives data from all remote nodes and compares the flow and pressure values.

Under normal conditions, the values remain consistent across all nodes. In case of leakage, there is a variation in flow rate and pressure between nodes. The system detects leakage based on these variations and identifies the affected section of the pipeline. This architecture ensures real-time monitoring, efficient data transmission, and accurate leakage detection in the water pipeline system.

III. METHODOLOGY

The methodology of the proposed system involves data collection, data processing, leakage detection, and communication. Each stage plays a vital role in ensuring accurate and real-time leakage detection.

A. Data Collection

The flow sensors and pressure sensors are used to continuously measure the water flow rate and pressure at different nodes in the pipeline. These sensors provide real-time data, which is collected by the ESP32 microcontroller at each node.

B. Data Processing

The ESP32 microcontroller processes the collected sensor data and converts it into meaningful values. The system compares the measured values with predefined threshold limits to identify abnormal variations in flow and pressure.

C. Leakage Detection

Leakage is detected by analyzing the difference in flow rate and pressure between different nodes. A significant variation in these parameters indicates the presence of leakage in the pipeline. The system determines the affected section based on these differences.

D. Communication

The processed data is transmitted from remote nodes to the central node using LoRa communication. The central node updates the real-time data on the Blynk mobile application and a web dashboard. This allows users to monitor the system remotely and take necessary actions.

IV. RESULTS

The proposed system was tested under real-time conditions to evaluate its performance in detecting water leakage. The system was initially observed under normal operating conditions, where the flow rate and pressure values remained stable across all nodes in the pipeline.



Fig. 1. Experimental setup of the proposed system

To analyze the system behavior during leakage, a controlled leak was introduced in the pipeline. A noticeable variation in flow rate and pressure values was observed between different nodes. The system successfully identified these variations and detected the presence of leakage.



Fig. 2. LCD display showing flow values

The location of the leakage was determined by comparing the sensor data from multiple nodes. The system accurately identified the affected section of the pipeline and estimated the leak distance from the central node.

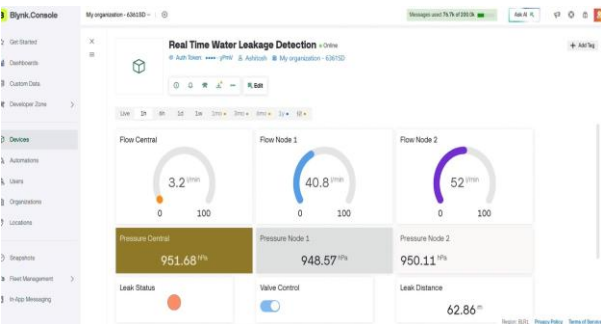


Fig. 3. Blynk dashboard showing real-time monitoring

The real-time data was displayed on multiple interfaces, including the LCD display, Blynk mobile application, and a web dashboard. The LCD provided local monitoring, while Blynk and the web dashboard enabled remote access to system data.



Fig. 4. Web dashboard showing leakage status

The experimental results demonstrate that the proposed system is capable of detecting leakage accurately with fast response time. The integration of IoT and LoRa communication ensures reliable data transmission and effective monitoring, making the system suitable for practical water management applications.

V. CONCLUSION

The proposed real-time water leakage detection system using IoT provides an efficient and reliable solution for monitoring pipeline conditions. By utilizing ESP32 microcontrollers along with flow and pressure sensors, the system continuously analyzes variations in pipeline parameters to detect leakage accurately. The integration of LoRa communication enables long-distance data transmission, while real-time monitoring through LCD display, Blynk application, and web dashboard enhances accessibility and control. The system also incorporates an automatic solenoid valve mechanism to prevent further water loss during leakage. Experimental results demonstrate that the system offers fast response, high accuracy, and reliable performance. Overall, the proposed solution is cost-effective and suitable for applications in smart cities, industries, and agricultural water management systems.

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